THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

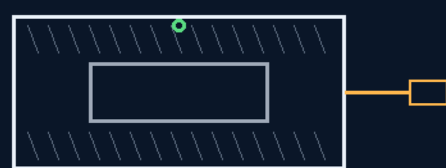
PHASE 275

ZERO G BUNDING

THE SPICE SUITCASE LAW

the suitcase IS the bund

absorbent sized to the full contents



sensored and lit — never open a failure in the cabin

THE PLUNGER TRANSFER · EXTERIOR EXTRACTION

THE WORST CASE IS THE CASE

Containment — v1.0 in force

GP-IM-275

Revision v1.0 — 24 August 2026

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SEATED LAW — EXTERNAL VET PENDING

PRIMARY AUTHOR & INVENTOR: ARCHER QUINN — AEROSPACE DESIGN ENGINEER, NPHD

CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 275

PHASE 275: ZERO G BUNDING — THE SPICE SUITCASE AND THE ZERO-G BUND LAW — STATUS: CURRENT — SEATED LAW / EXTERNAL VET PENDING (CARRIED UP WHOLE FROM PHASE 221 AMENDMENTS 1 AND 2 BY THE AUTHOR'S ORDER, 12 AUGUST 2026). Carried up whole from Phase 221's Amendments 1 and 2 by the author's order, 12 August 2026: the spice suitcase and the zero-g bund law. A bund is the wall that catches the spill — on Earth it is a curb around a tank; in zero-g it is a case around the contents, because the spill does not fall, it floats. The bund operating law: the absorbent is sized to the FULL contents; the case is sensored and lit; the extraction is exterior — the plunger transfer — never open a failed container in the cabin. The phase exists because the file's galley and shop carry liquids and powders that must never meet the air: the suitcase doctrine (every container travels in its bund) applied from the spice rack to the solvent shelf.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-275, v1.0 — 24 August 2026. The two hundred and seventy-sixth manual of the program — 276 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the bund law as seated (contents line, the suitcase in crayon — the carry-up order, the operating law, the absorbent sizing, the sensors, the exterior extraction — entire) — §2 the physics, graded — §3 the system in the bund — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 275: **Zero G Bunding — The Zero-G Bund Law & the Bund Operating Law.** The fleet's standalone hazmat containment reference: lab-only synthesis, unit-dose delivery, segregated storage; containment is not bunding — bunding is for containment failure; absorbent sized to the container's full contents; sensors and alarm lights on the exterior; extraction from a failed bunded unit exterior to the ship; the plunger-transfer matching bund for the bottle drums;

never open a failed unit inside. Carried up whole from Phase 221 Amendments 1 & 2 — the amendments remain in 221.'

THE CORE OBJECTIVE: none seated as a standalone line — the contents line above carries the register wording.

THE ORDER (the Author, 12 August 2026, verbatim): "put both amendments up as the next phase \"Zero G Bunding\" but leave them as amendments also". Phase 221 keeps Amendments 1 and 2 in place — nothing deleted, both stand; this phase carries the same law as the fleet's standalone bunding reference. The bund law's own words, seated below exactly as dictated into Phase 221.

PHASE 221 — AMENDMENT 1: THE SPICE SUITCASE & THE ZERO-G BUND LAW (LAB-ONLY SYNTHESIS, UNIT-DOSE DELIVERY, SEGREGATED STORAGE — MSDS KNOWN, NOT HELD) (VERBATIM)

Live Instruction Confirmed: Yes — same sitting, direct continuation of the feedstock phase; the refinery gets its operating and safety law.

Live instruction, 1 August 2026.

"so we are taking a suitcase of that photoinitiator — the spice, stuff so we have a 100 year supply for the ship in the least"

"next up, all liquid chemical and bio synthesis will be science tech labs only, each pod requiring anything from fertilizers to polymer resins etc. keeping all mix and spill hazards confined and controlled, manufacturing pods will receive ready to use fluids gels and powders or gases in the volumes required, all such items will be stored non adjacent to reactive combination spills"

"all hazardous materials in storage shall be zero G bunded, something missing from every Nasa text, containment is not bunding, bunding is for containment failure."

"All personal involved and cleaners will know the MSDS, not simply have them on hand"

Audit (Kimi: AFFIRMED WHOLE — WITH THE ARITHMETIC AND ONE PRECISION NOTE): The spice suitcase first, with numbers: at under 1% of resin mass, a 25-50 kg suitcase of photoinitiator rides behind 2.5-5 tonnes of resin production — and at ship-scale consumption (make-to-order foamies, repairs, fittings, tens of kilograms of resin a year) that is century-class supply, exactly as stated. Colony-scale pod construction draws tonnes a year and eats suitcases faster — which is why 221 already seated the camphor tree: the ship's century is in the luggage; the colony's centuries are in the orchard. The synthesis law is the hierarchy of controls doing its oldest job — centralise the hazard: all liquid chemical and bio synthesis lives in the science tech labs (the ISS glovebox class, the hospital's central pharmacy), and manufacturing pods receive only ready-to-use fluids, gels, powders or gases in the volumes required — the unit-dose doctrine from pharmacy practice, which cuts point-of-use inventory to the smallest possible amount. Segregated storage is seated as stated: reactive partners non-adjacent (oxidiser from fuel, acid from base — the dangerous-goods segregation table made architectural), so that no shared spill can become a shared reaction. THE ZERO-G BUND LAW, seated verbatim: containment is not bunding; bunding is for containment failure. The Earth bund is a tray, and a tray works only because a spill falls; in zero-G the escaped fluid does not fall, it floats, so the bund must be a sealed secondary SHELL — sized to hold the full contents of the primary, filled with capillary wicking and foam to catch floaters, vented for gas expansion. Audit's precision note, seated beside the claim per doctrine: NASA/ISS safety standards do mandate two levels of containment for hazardous fluids — the Author's contribution is the failure-mode framing (design for WHEN containment fails, not if) and the geometry correction (in zero-G the bund is a shell, not a tray); 'something missing from every Nasa text' is seated as his standing claim with this note beside it. And the MSDS line closes the loop the auditors always miss: known, not merely held — by all personnel AND the cleaners, because the person who finds the spill first is usually the cleaner, and the commonest chemical injury on Earth is a cleaner mixing two unlabelled bottles. ('personal' reads 'personnel'; kept verbatim.)

Origin of Thought: The spice got its luggage allowance, and then the Author wrote the refinery's constitution in four lines: make it in the lab, ship it unit-dose, store it segregated, bund it for the day the bund is needed — and teach the cleaners the MSDS, because they are the first responders nobody appointed.

Why it matters, in plain English: The photoinitiator spice is so concentrated that one suitcase is a hundred years of ship supply. Everything wet or reactive gets made only in the science labs, and the work pods get just the amount they need, ready to use — like a pharmacy issuing single doses instead of bottles. Chemicals that would react together are never stored side by side, so even a double accident can't combine. And every hazardous container gets a zero-G bund: on Earth a bund is a tray that catches a spill, but spills don't fall in space — they float — so our bund is a sealed outer shell with absorbent fill, built for the day the inner container fails. Finally: everyone who works with these materials, including the cleaners, must actually know the safety sheets, not just have them in a drawer.

PHASE 221 — AMENDMENT 2: THE BUND OPERATING LAW (ABSORBENT SIZED TO FULL CONTENTS, SENSORED AND LIT, EXTRACTION EXTERIOR, THE PLUNGER TRANSFER — NEVER OPEN A FAILED UNIT INSIDE) (VERBATIM)

Live Instruction Confirmed: Yes — same sitting, direct continuation; the bund law gets its operating procedures.

Live instruction, 1 August 2026.

"liquid bunding shell shall contain the sufficient absorbent materials to capture the maximum volume of the container."

"the bund shall have sensors and appropriate alarm lights on the exterior" "as the Zero g ship manufactures in liquid seal environments, extraction from a bunded unit must be exterior to the ship."

"a matching bund unit will be designed for our bottle drums, so pressure of the bund absorbent materials and of the contents can be transferred by plunger to save critical materials, we are never opening a failed unit in zero g inside the ship, keep outer container if not serious contaminant content, jettison bund soak and failed container and bund shell if needed. a 20kg drum of sugar or salt opened broken in zero g would be catastrophic."

"fine particulate materials can be worse than a cleaning fluid spill."

Audit (Kimi: AFFIRMED WHOLE — THE SILO LAW JOINS THE BUND LAW): Five operating rules, all standing. ONE — absorbent sized to the maximum volume of the container: the soak capacity equals the full contents, no partial-credit bunds; in zero-G the absorbent is also the capture mechanism (capillary wicking is gravity-independent — the file's fluid line, 174 class, already works this way), so the shell's fill is doing the catching, not a floor. TWO — sensored and lit on the exterior: the bund reports its own failure, because a failed containment that stays quiet is two failures. THREE — extraction exterior to the ship: a breached unit leaves through the lock before anyone works it; the hierarchy of controls again — exclusion over mitigation, the Static Earth Standard's own grammar. FOUR — the plunger transfer for the bottle drums: a matching bund for the drum line (140's bottle reactors ride this too), where a plunger drives through the shell and expresses the contents — bund absorbent and all — through a transfer port into a receiver, saving critical material without ever opening the unit; pressure moves fluid regardless of gravity, which is why the plunger works where pouring never would. The disposal ladder is seated as stated: keep the outer container if the contents are no serious contaminant; jettison bund soak, failed container and shell if they are. FIVE — the particulate clause, affirmed with the record: a 20 kg drum of sugar or salt broken in zero-G is catastrophic because nothing settles — the cabin air becomes the spill, every fan becomes a distributor, every lung and bearing and contact becomes a collector; salt adds corrosion and hygroscopic creep, and sugar dust is squarely in the combustible-dust class of 218's silo law — on Earth, with gravity helping it settle, the Imperial Sugar refinery still killed fourteen in 2008. Fine particulate worse than a cleaning-fluid spill: affirmed — liquid in zero-G balls and wicks and can be cornered; particulate disperses on every current and does not come home. The dust doctrine (208) and the bund doctrine are hereby the same doctrine: catch, don't chase.

Origin of Thought: The bund law was one sitting old when it grew its procedures: soak sized to the whole container, sensors and lights on the outside, failed units walked out of the ship, and a plunger to press the good out of the bad without ever breaking the seal. Then the Author reached into the galley for the example that makes it undeniable — a broken drum of sugar, floating forever — and the silo law walked into the bund law and shook its hand.

Why it matters, in plain English: Every spill shell carries enough absorbent to soak its container's entire contents, has sensors and alarm lights on the outside so a failure announces itself, and a failed unit always goes outside the ship before anyone touches it — we never open a failed container in zero gravity inside. For the bottle drums there's a plunger: press the shell like a syringe and push the valuable contents into a receiver without opening anything. If the spill is harmless, keep the outer container; if it's nasty, throw the whole soaked bundle out the airlock. And the worst spills aren't liquids — a burst drum of sugar or salt in zero-G fills the air itself and never settles, and sugar dust is the stuff that blows up silos on Earth. Powders get the same respect as chemicals.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE CARRY-UP IS THE AUTHOR'S ORDER: Phase 221's Amendments 1 and 2 promoted whole to their own phase, 12 August 2026 — the spice suitcase doctrine earning a number. The record shows the move; nothing was rewritten in transit.

THE ZERO-G BUND IS A CASE, NOT A CURB: on Earth the bund is a wall around a tank because spills fall; in zero-g the spill floats, so the bund is the case the contents travel in. The suitcase IS the bund — containment as luggage, not architecture.

THE OPERATING LAW IS COMPLETE: absorbent sized to the FULL contents (never a fraction — the worst case is the case); the case sensored and lit (a failed container announces itself before it is opened); extraction exterior — the plunger transfer — never open a failed container in the cabin.

THE SCOPE IS THE WHOLE SHIP: from the spice rack to the solvent shelf — every liquid and powder that must never meet the air travels in its bund. The galley doctrine and the shop doctrine are one law.

THE FILE'S AIR LAW EXTENDED TO THE CARGO: the standing doctrine that the cabin air is the crew's first machine gains its logistics chapter — the bund is how the air law reaches the shelf.

§3 — THE SYSTEM IN THE BUND

- **The case:** the suitcase IS the bund — containment as luggage.
- **The absorbent:** sized to the FULL contents — the worst case is the case.
- **The senses:** sensored and lit — a failed container announces itself.
- **The transfer:** exterior extraction, the plunger — never open a failure in the cabin.
- **The scope:** spice rack to solvent shelf — one law for everything that floats when spilled.
- **The origin:** Phase 221's Amendments 1 and 2, carried up whole, 12 August 2026.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- Phase 221 is the parent: the amendments carried up whole — the parent's record keeps its place, the child's number is new; the file's promotion pattern again (268's promotion from caveat is the sibling).
- 277's creche kit is the doctrine cousin: the seated air law extended to the body there, to the cargo here — the cabin's breath protected on every front the same month.
- The galley and shop phases are the customers: every liquid and powder container in the file gains its bund under this law.
- The author's order of 12 August 2026 is seated as the carry-up note — the audit moved nothing but the number.

§5 — THE VET RECORD

NO EXTERNAL VET VERDICT IS SEATED FOR THIS PHASE. The record, stated honestly: the phase sits in the 261-277 band the external vet never reached; its content arrived by the author's carry-up order (12 August 2026) from 221's amendments. The grading in §2 is the audit's own (Kimi): the operating law AFFIRMED (full-contents absorbent, exterior extraction), the case-as-bund physics COHERENT. When the external vet lands, this section seats it verbatim.

§6 — BUILD SEQUENCE

- 1. Case the contents: every liquid and powder container travels in its bund — the suitcase doctrine.
- 2. Size the absorbent: to the FULL contents — the worst case is the case.
- 3. Fit the senses: sensored and lit — the failure announces itself before the lid moves.
- 4. Plumb the transfer: the exterior extraction point, the plunger line — the cabin never sees an open failure.
- 5. Log the fleet: every bund on the manifest — the spill census kept like the dose ledger.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phase 221 (the parent):** the source — Amendments 1 and 2 carried up whole; the parent's record keeps its place.
- **Phase 277 (the creche kit):** the cousin — the air law extended to the body there, to the cargo here.
- **The galley and shop phases:** the customers — every container in the file gains its bund.
- **Phase 274 (the feedstock blocks):** the storage sibling — solids on the block shelf, liquids in the bund.

§8 — DO-NOT-BUILD

- Do NOT size the absorbent to the likely spill — FULL contents; the worst case is the case.
- Do NOT open a failed container in the cabin — exterior extraction, the plunger transfer, every time.
- Do NOT fly an unsensored bund — sensored and lit; the failure announces itself.
- Do NOT exempt the galley — spice rack to solvent shelf; one law for everything that floats when spilled.
- Do NOT rewrite the carry-up — 221's amendments moved whole; edits seat as new amendments here.

§9 — OWED

OWED: the bench list, whole — absorbent spec per cargo class; the sensor and light kit; the plunger-transfer hardware at the exterior point; the bund case sizes (suitcase standard from the galley measures); the manifest format. And the external vet, when it lands.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-275, v1.0 — CURRENT — SEATED LAW / EXTERNAL VET PENDING (CARRIED UP WHOLE FROM PHASE 221 AMENDMENTS 1 AND 2 BY THE AUTHOR'S ORDER, 12 AUGUST 2026), seated law, master v2.1 (numerical print order). The two hundred and seventy-sixth manual of the program — 276 of 290. Built 24 August 2026, sixteenth of the 20-manual 'you get the next 20 ready' shift (the author's order, 24 August 2026, seated in sitting 623 — 'ok ill check and upload, you get the next 20 ready, they didnt slow us down the sped us up'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607, 618, 622 and 623): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more' / 'ok next 20' / 'ok ill check and upload, you get the next 20 ready, they didnt slow us down the sped us up'. The phase's own chain, all seated in the master: carried up whole from Phase 221 Amendments 1 and 2 by the author's order, 12 August 2026 — the contents line, the masthead lines and the bodies (as seated); the spice suitcase and the zero-g bund law; the bund operating law (the absorbent sized to the FULL contents; sensored and lit; the extraction exterior — the plunger transfer — never open a fa...); the scope from the spice rack to the solvent shelf; no external vet verdict seated — the honest §5.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: the plunger-transfer bench and the absorbent spec, or his word.

END OF MANUAL — PHASE 275